CSCI 220/620-131: Discrete Structures
Course Syllabus

Instructor Contact Information

Instructor: Kathy He
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Office: Science Building A201
Office Hours: Mondays, 3:45pm-4:45pm

Course Description

(*CSCI 220's official description on QC website*)
3 lec. hr.; 3 cr.
Algorithms, recursion, recurrences, asymptotics, relations, graphs and trees, applications.
(CSCI 620's official description on QC website)
3 hr.; 3 cr.
Algorithms, recursion, counting problems, recurrences and their closedform solutions, asymptotic analysis of functions, relations, graphs and trees, applications to computational problems.

The following is a general outline of the topics we will cover, in the likely order we will cover them (along with corresponding textbook sections):

- Propositional logic and equivalences (Chapters 1.1 and 1.3)
- Growth of functions (Chapters 3.2 and 3.3)
- Induction and recursion (Chapters 5.1 and 5.2)
- The Pigeon Hole Principle (Chapter 6.2)
- Recurrence relations (Chapters 8.1-8.3)
- Relations (Chapters 9.1 and 9.5)
- Graphs (Chapters 10.1-10.4, 10.6, and 10.8)
- Trees (Chapters 11.1-11.4)
- Discrete probability, probability theory, Bayes' Theorem, expected value and variance (Chapters 7.1-7.4)

Prerequisites

- MATH 120 (or 220)
- MATH 141 or 151
- CSCI 111 (or 112)

Required Textbook

Rosen, K.H. (2018). *Discrete Mathematics and Its Applications* (8th ed.). McGraw Hill.

Homework

- Students should expect to spend, on average, about 5 to 6 hours per week on homework for this class.
- Homeworks will consist of readings and exercises from the Rosen textbook, assigned each class.
- Assignments will be regularly posted on Blackboard.
- Homework is ungraded. It is your responsibility to determine that:
    - You have done the homework problems correctly, and
    - You also understand why your solutions are correct
    - If in doubt, discuss with classmates, make an appointment at the Learning Commons, or come to my office hours.
- The midterms and the final will be **difficult to pass** without attending classes, doing and understanding the homework!

Course Schedule and Important Dates

All lectures will be held in person, Mon./Weds. 5:00pm – 6:15pm in Powdermaker Hall Room 116. Any room changes will be announced on Brightspace (and corrected on syllabus).

- Monday, 8/31—First lecture
- Monday, 9/7—No class (College closed)
- Monday, 9/21—No classes scheduled
- **Monday, 10/5—Midterm #1**
- Monday, 10/12—No class (College closed)
- Tuesday, 10/13—We will have class; college follows a Monday schedule
- **Wednesday, 11/4—Midterm #2**
- Friday, 11/6—Last day to drop with a grade of W
- Wednesday, 11/25—No classes scheduled
- **Monday, 12/7—Midterm #3**
- Monday, 12/14—Review for Final
- **Wednesday, 12/16—Final Exam, 4:00pm-6:00pm held in PH 113**

Final exam scheduling is done by the college and may be subject to change. Check CUNYfirst regularly. I will also make an announcement in class and on Brightspace if there are any changes.

Grade Calculation

There will be three midterms and one cumulative written final for undergraduate students. For graduate students, there will be an additional individual oral final exam. The scores for these examinations are weighted as follows:

Undergraduate Students (CSCI 220)		Graduate Students (CSCI 620)	
Midterm #1:	20%	Midterm #1:	20%
Midterm #2:	20%	Midterm #2:	20%
Midterm #3:	20%	Midterm #3:	20%
Cumulative Written Final:	40%	Cumulative Written Final:	30%
		Oral Final Exam:	10%

The lowest of the three midterm scores will be replaced by the cumulative written final exam score if doing so results in a higher numeric score. Depending on the distribution of scores at the end of the semester, a curve may be applied when calculating final grades.

Course Policies

Announcements

- Announcements will be made in the beginning of class, as well as on Brightspace.
- I advise students to double check their Notifications settings on Brightspace (Found in a dropdown menu by clicking your name on the top right corner) to ensure that they are receiving emails for new announcements.

Absence Policy and Emergencies

- Attendance will be taken in the beginning of class. Latecomers must come to me after class to have their attendance recorded.
- In the case of an absence, students are responsible for obtaining notes from a classmate to make up the material.
- In the unfortunate event of a **valid and documented emergency**, e.g., a positive COVID test, missing an exam will be handled via a one-on-one oral examination to be conducted in person.

- **If you are absent for the first five lectures, you will receive a grade of WN for the course.**

Academic Integrity

- Any student who engages in any form of academic dishonesty on
  - the final exam will receive an automatic F as their overall course grade.
  - the midterm will receive a 0 for the midterm and one's overall numerical grade will be dropped by 6 points.
- Academic dishonesty is prohibited in The City University of New York. Penalties for academic dishonesty include academic sanctions, such as failing or otherwise reduced grades, and/or disciplinary sanctions, including suspension or expulsion. Incidents will be handled in accordance with CUNY's Academic Integrity Policy. Please read this policy and familiarize yourself with what constitutes academic dishonesty and the consequences for violating this policy.

Communication Policy

- Ensure that your preferred email is the official Queens College issued email provided by the school, and that this is the email address you use to send emails to the instructor's email listed in this syllabus (kathy.he30@qc.cuny.edu). Emails sent from non-CUNY domains will be ignored.
- Before emailing the instructor, consult this syllabus first to ensure your issue has not already been addressed in the syllabus.
- If a matter requires lengthy discussion and you can not make it to office hours, please email me to schedule a separate appointment.

Grading

- All grades are final and will not be changed except in the rare event of an error or miscalculation on my part.
- A grade of "Incomplete" will only be given in cases where it applies per QC/CUNY policy.

Exams

- No make-up exams will be given. Missing an exam results in a 0 for that exam.
- **Missing the final will result in a WU for the course.**
- To take any of the midterms for this course, as well as the final, students must provide either their **Queens College ID** or a governmental picture ID to verify their identity.
- Students may not leave the room until **after** their midterm/final has been handed in.

- Unless stated otherwise, all exams are closed-book, closed-note, and ALL electronic devices (phones, laptops, tablets, smart watches, smart glasses, headphones, etc.) must be packed away and inaccessible. Failure to adhere to these exam policies will result in a F for that exam and will be reported as academic dishonesty.

Campus Resources

Reasonable Accommodations For Students With Disabilities

For students with disabilities needing academic accommodation:

- Register with the Special Services Office by emailing: QC.SPSV@qc.cuny.edu.
- Provide me with a letter from the Special Services Office within the first week of class.
- For more info, see: https://www.qc.cuny.edu/sp/

Counseling Services

The mission of Counseling Services is to enhance students' academic, intellectual, personal, and social growth. Special attention is given to students' health and well-being, with the aim of alleviating the effects of painful experiences, enhancing self-understanding and understanding of others, and fostering students' pursuit of their goals. Counseling Services also presents workshops, training, and educational consultation to the college community.

For more on the resources available to Queens College students, see: https://www.qc.cuny.edu/cs/

On-Campus Peer Tutoring and Study Spaces

The Queens College Learning Commons (QCLC) will be open for tutoring starting Monday, 9/14/2026 in Kiely Hall, Room 131. QCLC offers free tutoring, workshops, and study resources to support student success across disciplines.

Tutoring Hours:
Days: Monday - Thursday
Time: 10:00AM - 6:00PM
Location: Kiely Hall, Room 131
Mode: In-person and virtual tutoring via zoom

👉 Make an appointment: https://qclc.mywconline.com/

🌐 Learn more: https://www.qc.cuny.edu/academics/qclc/
✉️ Contact: LCommons@qc.cuny.edu

Acknowledgement

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